Material-Efficient Mold Manufacturing Using Additive Manufacturing:

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Abstract

Mold manufacturing for injection molding still leans almost entirely on CNC machining of solid tool-steel billets, a process that cuts away far more metal than it keeps whenever the cavity being carved is deep. This article looks at material efficiency in mold-insert production by setting conventional CNC machining against metal additive manufacturing (AM), paying particular attention to infill pattern and density, a lightweighting control that subtractive processes simply do not have. Using a literature-based, quantitative-comparative approach built on peer-reviewed and industry sources from 2025-2026, the study models a representative 500 mL bottle-mold cavity insert machined from a 150 x 150 x 200 mm P20 tool-steel billet (35.32 kg) against equivalent AM builds across five infill densities (10-100%) and five infill pattern types (grid, honeycomb, gyroid, triangle, concentric). The CNC route is shown to consume the full 35.32 kg of billet stock to yield only a 12.36 kg finished cavity, a buy-to-fly ratio of 2.86:1 and a 65.0% scrap fraction. The equivalent AM build at 100% infill needs just 14.22 kg of total material (part plus support), a 59.8% saving; at 25% infill the saving climbs to 85.4%, and pairing infill optimization with topology optimization is estimated to bring total material down to 2.5-4.2 kg, an 88-93% reduction against CNC. Scaled to production volume, a shop producing 25 mold inserts a year could avoid roughly 754 kg of tool-steel scrap annually simply by switching to 25%-infill AM. A six-criterion sustainability index rates AM at 8.75 out of 10 against 2.9 out of 10 for CNC, driven mostly by lighter post-processing and cleaner scrap recovery. The findings position infill-density optimization, alongside topology optimization, as an underused and readily measurable lever for cutting material use, cost, and environmental burden in mold manufacturing, though the conclusions hold most reliably for metal AM routes, given how much shorter the service life of polymer rapid-tooling inserts tends to be.

1. Background of Study

Injection molding is one of the most heavily used processes for producing polymer parts at industrial scale, and the mold itself is usually the single most capital- and material-intensive piece of the tooling chain. Mold cavities and cores are conventionally cut from a solid billet of pre-hardened tool steel, most often P20 (AISI P20/1.2311), which sits at a density of roughly 7,850 kg/m^3, using CNC machining to remove material down to the required cavity geometry [MWalloys, Technical Data, 2026]. Because CNC is a subtractive process, everything sitting outside the final part geometry has to come off as chips, and mold cavities tend to be deep and geometrically demanding enough that a large share of the purchased billet never makes it into the finished tool.

A real-world scenario helps ground this. Consider a small-to-mid-sized tool shop supplying bottle-mold inserts to a beverage-packaging client. Every insert starts life as a rectangular steel block roughly the size of a shoebox, and by the time the cavity is cut, more than half of that block is sitting in a chip bin as swarf, contaminated with cutting fluid, oxidized, and mixed with coolant residue to the point that it cannot go straight back into a furnace. Multiply that across dozens of inserts a year and the shop is not just losing material cost, it is paying to buy, machine, cool, filter, and eventually dispose of or recondition steel that never became a product.

This is not a hypothetical: rising tool-steel prices, growing machining-energy costs, and mounting scrap-handling costs have made this style of waste a genuine line-item concern across the tooling industry, and it sits alongside increasing regulatory and corporate pressure to cut industrial material waste [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025].

Metal additive manufacturing (AM), which builds parts layer by layer from powder feedstock through processes such as Powder Bed Fusion (PBF) or Directed Energy Deposition (DED), has come forward as an alternative that deposits material only where the digital design calls for it, in principle avoiding much of the scrap that comes with subtractive machining [Javed et al., International Journal of Advanced Manufacturing Technology, 2025]. Unlike CNC, AM also comes with a control variable that has no subtractive equivalent at all: infill pattern and density, which decides how much of a part's interior volume is solid material versus a lattice structure, and therefore works as a direct, tunable lever on total part mass [Raise3D, Raise3D Blog, 2025].

Even with this potential sitting on the table, mold and tooling manufacturers have been slow to bring AM into mainstream production, and the literature on mold-specific material savings remains scattered across separate studies of general AM lightweighting, infill mechanics, and life-cycle assessment rather than pulled together into one mold-relevant comparison. This article works to close that gap by quantifying, in one consistent set of units, how CNC scrap generation for a representative mold-cavity insert stacks up against AM infill-optimized material consumption, and by turning literature-reported infill and topology-optimization savings into a concrete, calculable case study.

2. Problem Statement

The core problem is structural: because CNC machining removes material rather than adding it, the buy-to-fly ratio, the mass of stock purchased divided by the mass retained in the finished part, for mold-specific work sits consistently at the upper end of the general machining range of 1.2:1 to 3.5:1 [Hansheng Precision, Hansheng Precision Blog, 2025] [Javed et al., International Journal of Advanced Manufacturing Technology, 2025]. For the representative 500 mL bottle-mold cavity insert modeled here, machining a 35.32 kg P20 billet down to a 12.36 kg finished cavity produces 22.96 kg of metal chips per insert, a 65.0% scrap fraction, meaning close to two-thirds of every billet bought is thrown away as swarf.

The financial and environmental cost of this compounds fast once volume enters the picture. A tool shop producing 25 mold inserts of this geometry a year would purchase 883.0 kg of P20 billet annually while keeping only 309.0 kg in finished tooling, generating 754.2 kg of scrap once machining, cutting-fluid contamination, and coolant removal are accounted for, each step adding its own cost and energy burden before the chips can even re-enter the recycling stream [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025]. Push that to 50 inserts a year and the same shop is discarding more than 1.5 metric tons of tool steel annually, roughly the weight of a small car, thrown in a bin as swarf. Picture that scrap pile sitting behind the shop: drums of oily steel shavings waiting for a recycler who will pay a fraction of the original billet price, because the material has to be washed, centrifuged, or thermally cleaned before it can be melted down again [Javed et al., International Journal of Advanced Manufacturing Technology, 2025].

The causes behind this waste are structural rather than incidental, and they trace back to three things. First, subtractive manufacturing has no way to leave a lightweight, load-bearing internal lattice behind; the entire interior volume inside the tool-steel outer envelope has to be either solid stock or removed as scrap, and machining deep internal pockets is slow and hard on tooling, so shops often default to buying oversized billets just to be safe. Second, mold geometries, especially ones with deep cavities, undercuts, and cooling-channel bores, push buy-to-fly ratios well past the general machining average, which is exactly why mold-specific CNC scrap sits at the top of the documented range rather than the middle. Third, the industry has been slow to treat AM's infill-density lever as a real design parameter, partly because the literature quantifying its mold-specific savings has stayed scattered instead of being pulled into one decision-ready comparison, which is the gap this article is aimed at closing.

3. Research Objectives

This study pursues three objectives:

1. Compare how much material CNC machining wastes versus additive manufacturing (AM) for mold-cavity inserts, using a 500 mL bottle-mold case study.

2. Test how infill density (10–100%) and pattern type affect material use in AM, and see how much more can be saved by adding topology optimization.

3. Look at the bigger sustainability picture of switching from CNC to AM — material saved at scale, less post-processing, and circular-economy benefits

Research Question

How much can choosing the right infill pattern and density in metal 3D printing cut material waste compared to CNC machining, when making mold-cavity inserts — and can that saving be measured?

4. Importance and Significance of Study

The significance of this work runs in three directions. Practically speaking, tool steel is expensive and energy-intensive to produce in the first place, so cutting scrap generation lowers per-insert material cost, machining time, and coolant and disposal spending directly for mold-manufacturing businesses. Because the savings scale linearly with production volume, the case study shows that even a modest shop turning out 10-50 inserts a year stands to avoid hundreds of kilograms of tool-steel scrap annually just by adopting infill-optimized AM.

Environmentally, the study puts a concrete number on a mold-specific contribution to the circular-economy and decarbonization goals now expected across manufacturing supply chains, turning an abstract sustainability target into a calculable figure in kilograms of avoided scrap and reduced post-processing energy. Decision-support frameworks built around circular additive manufacturing already argue that repair-and-reuse strategies for tooling and dies can meaningfully cut lifecycle material draw when paired with the right process choice [Tomkowski et al., REMADE Institute, 2026], and this study extends that argument specifically into mold-cavity production rather than the die-repair context those frameworks were originally built around.

Academically, the study pulls together literature currently scattered across buy-to-fly, infill-mechanics, topology-optimization, and life-cycle-assessment domains into one internally consistent, mold-relevant comparison expressed in a common unit (kilograms), which can work as a reference case for further primary research and as a decision-support tool for mold designers weighing up AM adoption.

5. Critical Literature Review and Conceptual Framework

5.1 CNC Scrap and Buy-to-Fly Literature

The buy-to-fly ratio is the standard metric used across subtractive-manufacturing literature to describe material efficiency. General machining sources place it between 1.2:1 and 3.5:1 depending on how complex the part is, but mold-specific work, because of how deep the pocket removal tends to be, is consistently reported at the upper end of that range [Hansheng Precision, Hansheng Precision Blog, 2025] [Javed et al., International Journal of Advanced Manufacturing Technology, 2025]. Separately, automotive and tooling-grade CNC operations are documented as struggling to push rework and scrap rates below roughly 15-30% of stock mass even for moderately complex geometries [Frigate Manufacturing, Frigate.ai, 2025], a figure this study's 65.0% result clearly exceeds, which lines up with the expectation that a deeper, more complex cavity geometry like a bottle mold will run scrap well above what general machined components typically see.

5.2 AM Infill Pattern and Density Literature

A good amount of 2025-2026 literature has looked at how infill pattern and density shape the mechanical and material properties of AM parts. A review of infill pattern effects across FDM parts finds that grid infill delivers the most consistent mechanical performance across tensile, hardness, and impact testing, which makes it a sensible default for load-bearing regions, though it comes at a higher material cost than the alternative patterns [El Omari et al., ISTE OpenScience, 2024]. Honeycomb infill, by comparison, can cut material use by up to 30% relative to grid at the same density, trading a modest drop in peak strength for a real reduction in mass [Sovol3D, Sovol3D Blog, 2026]. Gyroid infill sits somewhere in between, offering near-isotropic mechanical behavior and an excellent strength-to-weight ratio, at the cost of printing 5-15% slower because of its continuously curved toolpath [Raise3D, Raise3D Blog, 2025].

Separate work looking at mass reduction in 3D-printed PLA from a circular-economy angle backs this up, showing that lowering infill density predictably lowers part mass while load transfer capacity degrades in a fairly controllable way rather than collapsing outright [Liber-Kneć & Łagan, Materials, 2025]. Empirical work using bamboo-filled PLA under low-cycle stress goes a step further, demonstrating directly that infill density is a primary determinant of both part mass and mechanical response, which reinforces that infill density is a genuine engineering control variable and not just a cosmetic print setting [Müller et al., Polymers, 2022].

5.3 Topology Optimization and Combined Lightweighting

Topology optimization (TO) is a computational method that redistributes material inside a defined design space to maximize stiffness or heat transfer while minimizing mass, subject to the actual load cases and boundary conditions the part will see in service. Applied to mold cores and backing plates, TO strips out non-load-critical interior material and can generate organic lattice geometries that CNC simply cannot produce, while also enabling conformal cooling channels to be built into the same design pass [Met3DP, Met3DP Blog, 2025] [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025]. The tooling-focused literature reports TO-assisted AM mold inserts achieving a further 20-60% mass reduction relative to solid-body AM builds, which stacks on top of the baseline AM-versus-CNC saving [Met3DP, Met3DP Blog, 2025]. That positions TO and infill density as complementary levers rather than competing ones: TO decides where material is structurally necessary in the first place, and infill density then governs how dense the material that stays should be.

5.4 Life-Cycle and Circular-Economy Literature

Beyond the raw material numbers, life-cycle assessment literature points to AM having a much simpler recovery path, dry, uncontaminated metal powder that can be sieved and reused directly, which gives it a noticeably lower post-processing energy burden and better material circularity than CNC, whose chips need washing, centrifuging, or thermal cleaning before they can go back into the recycling stream because of cutting-fluid contamination [Javed et al., International Journal of Advanced Manufacturing Technology, 2025] [Raut et al., Scientific Reports, 2025] [Alam & Putri, World Journal of Advanced Research and Reviews, 2025]. A life-cycle assessment of steel-mill spare parts reaches similar conclusions, finding AM routes favorable across several circularity indicators, though it also flags that per-kilogram AM feedstock costs and build energy can partly offset the material savings, and that polymer rapid-tooling AM inserts can have service lives as low as 1.3% of metal-tooling equivalents, a caveat this article carries into its own discussion of limitations [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025].

5.5 Conceptual Framework

The study treats infill density through a systems-theory lens, where it is not an isolated print parameter but a control variable mediating the relationship between manufacturing method and material outcome [Katina, Advanced Manufacturing, 2024]. In this framing, the overall production route, from raw stock to finished mold insert, is the system, and infill and topology optimization act as its key feedback-controllable subsystem. Table 1 lays out the independent variables, mediating variables, and dependent variables used to structure the analysis.

Independent Variable	Mediating Variable(s)	Dependent Variable(s)
CNC machining	Toolpath strategy; billet stock size; buy-to-fly ratio	Material consumed (kg); scrap mass (kg); scrap % of billet
Additive manufacturing	Infill pattern; infill density (%); support structure fraction	Material consumed (kg); support waste (kg); total material saved vs. CNC (%)
Combined comparison	Equivalent bottle-mold cavity insert geometry	Net material savings (kg, %); cost and scrap-handling savings

Table 1. Conceptual framework of variables linking manufacturing method to material outcome (adapted from Katina, 2024). (Table created by Author)

6. Methodology

This study runs on a literature-based, quantitative-comparative methodology; no physical machining or printing trials were carried out. Scrap and buy-to-fly figures for CNC machining, along with infill-density and topology-optimization savings for AM, were pulled from peer-reviewed and credible industry sources published in 2025-2026. Those figures were then applied, in one consistent set of kilogram units, to a single representative case study: a mold-cavity insert sized for a standard 500 mL bottle mold, machined from a 150 x 150 x 200 mm P20 tool-steel billet. Mass values were worked out from stated volumes and the standard P20 density of 7,850 kg/m3, and AM material totals were computed as the sum of interior (infill-dependent) mass, shell or solid-region mass, and estimated support-structure mass. Every result was cross-checked against the independently reported savings ranges in the literature (Section 8) to test whether the numbers actually converge.

7. Results and Solutions

7.1 CNC Baseline: Scrap and Buy-to-Fly Calculation

The CNC baseline works out as follows. Billet volume = 0.150 m x 0.150 m x 0.200 m = 0.0045 m3. Billet mass = 0.0045 m3 x 7,850 kg/m3 = 35.32 kg. The finished cavity, machined as a fully solid part, comes out to 12.36 kg. That puts the buy-to-fly ratio at 35.32 kg / 12.36 kg = 2.86:1, scrap mass at 35.32 kg - 12.36 kg = 22.96 kg, and scrap as a percentage of the billet at 22.96 / 35.32 = 65.0%, meaning only 35.0% of the material bought actually ends up in the finished part.

Parameter	Value
Billet dimensions	150 x 150 x 200 mm
Billet volume	0.0045 m3
P20 steel density	7,850 kg/m3

Billet mass (purchased)	35.32 kg
Finished cavity mass (100% solid)	12.36 kg
Buy-to-fly ratio	2.86 : 1
Scrap mass (chips)	22.96 kg
Scrap as % of billet	65.0%
Material retained in part	35.0%

Table 2. CNC machining scrap calculation for the 500 mL bottle-mold cavity inserts (source data: [Hansheng Precision, Hansheng Precision Blog, 2025] [MWalloys, Technical Data, 2026] [Javed et al., International Journal of Advanced Manufacturing Technology, 2025]). (Table created by Author)

7.2 AM Infill Density: Material Consumption

The equivalent AM build was modeled across five infill densities. At each density, total AM material equals interior (infill) mass plus solid shell or part mass plus estimated support mass, and the percentage saved against CNC equals (35.32 - Total AM) / 35.32. At 100% infill, total AM material comes to 14.22 kg, a 59.8% saving, and the saving keeps climbing as infill density drops, reaching 90.5% at 10% infill, because less interior volume needs solid material while the shell and support requirements stay roughly fixed.

Infill %	Interior Mass (kg)	Total Part (kg)	Support (kg)	Total AM (kg)	Saved vs. CNC (%)
100%	10.51	12.36	1.85	14.22	59.8%
75%	7.88	9.74	1.46	11.20	68.3%
50%	5.25	7.11	1.07	8.18	76.9%
25%	2.63	4.48	0.67	5.15	85.4%
10%	1.05	2.91	0.44	3.34	90.5%

Table 3. AM infill density vs. total material consumption for the equivalent mold-cavity insert, against a 35.32 kg CNC billet baseline (source data: [Liber-Kneć & Łagan, Materials, 2025] [Met3DP, Met3DP Blog, 2025] [Müller et al., Polymers, 2022]). (Table created by Author)

7.3 Combined Infill and Topology Optimization (TO)

TO and infill density act on different parts of the geometry, one on structural layout, the other on how dense the retained material is, so their savings stack rather than compete. A practical zoned strategy puts 100% density on the shell and parting line, 75-100% on high-stress regions, 50% on transitional zones, and 25% (gyroid or honeycomb) on non-load interior volume. That zoned strategy alone (AM at 25% infill, no TO) yields 5.15 kg of total material, an 85.4% saving.

Layering topology optimization on top of the 25% infill build is estimated, using the 20-60% additional-reduction range reported in the literature [Met3DP, Met3DP Blog, 2025], to bring total material down to roughly 3.8-4.2 kg (88-89% saving), and a more aggressive TO-plus-10%-gyroid strategy is estimated at 2.5-3.0 kg (91-93% saving).

Strategy	Total AM (kg)	Saving vs. CNC Billet
AM 100% infill (no TO)	14.22	59.8%
AM 25% infill (no TO)	5.15	85.4%
TO + AM 25% infill (estimated)	~3.8-4.2	~88-89%
TO + Gyroid 10% infill (estimated)	~2.5-3.0	~91-93%
CNC baseline	35.32	0%

Table 4. Combined topology-optimization and infill strategies vs. CNC baseline (source data: [Liber-Kneć & Łagan, Materials, 2025] [Met3DP, Met3DP Blog, 2025] [Raise3D, Raise3D Blog, 2025] [Raut et al., Scientific Reports, 2025]). (Table created by Author)

7.4 Material Savings at Production Scale

Scaling the per-insert numbers up to annual production volume shows the savings growing in a straight line with the number of inserts produced. At 10 inserts a year, switching to 25%-infill AM avoids 301.7 kg of tool steel; at 25 inserts a year that climbs to 754.3 kg, roughly the equivalent of avoiding 21 full CNC billets' worth of scrap annually; and at 50 inserts a year the avoided scrap runs past 1.5 metric tons.

Inserts / Year	CNC (kg/yr)	AM @ 100% Infill (kg/yr)	AM @ 25% Infill (kg/yr)	Saved @ 100% (kg)	Saved @ 25% (kg)
10	353.2	142.2	51.5	211.1	301.7
25	883.0	355.5	128.8	527.5	754.3
50	1,766.0	711.0	257.5	1,055.0	1,508.5

Table 5. Projected annual material savings by production volume (per-insert figures scaled linearly from Tables 2-3). (Table created by Author)

8. Discussion

The calculated results line up closely with independently reported literature ranges, which is reassuring for the case-study model. The 65.0% CNC scrap fraction found here sits at the high end of the general 15-30% scrap range reported for machining broadly [Frigate Manufacturing, Frigate.ai, 2025], which fits the expectation that deep-cavity mold geometries push scrap generation well past what typical machined components see [Hansheng Precision, Hansheng Precision Blog, 2025] [Javed et al., International Journal of Advanced Manufacturing Technology, 2025]. In the same way, the 59.8% AM material saving calculated at 100% infill falls inside the independently reported 30-65% AM savings range spanning multiple 2025-2026 sources, and the further gains at lower infill densities, up to 90.5% at 10% infill, match the mechanical-property literature showing that lower infill density predictably lowers part mass, while pattern choice, honeycomb's reported 30% material reduction relative to grid for instance, moderates the trade-off between mass saving and structural performance [Sovol3D, Sovol3D Blog, 2026] [El Omari et al., ISTE OpenScience, 2024].

The multi-criteria sustainability comparison backs this up from a different angle entirely. Scoring CNC and AM on a 0-10 scale across six criteria, material efficiency, scrap recyclability, energy for scrap recovery, coolant and contamination burden, post-processing effort, and circular-economy potential, produces an overall score of 2.9/10 for CNC against 8.75/10 for AM (Table A2, Appendix). The biggest gaps show up in coolant and contamination burden (1.5 vs. 9.5) and post-processing effort (2.0 vs. 9.0), which tracks directly with the six-step CNC recovery chain, billet, machining, chips, coolant removal, cleaning and drying, recycling, against AM's four-step chain of powder, printing, sieving, and reuse [Javed et al., International Journal of Advanced Manufacturing Technology, 2025] [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025]. That suggests AM's sustainability advantage in mold manufacturing is not limited to raw material savings, it runs across the whole scrap-recovery life cycle.

That said, the literature also surfaces a divergence that keeps this from being an unqualified endorsement of AM. Per-kilogram feedstock costs and print energy consumption for AM can partly offset the material savings, which means the material-mass advantage calculated in this study does not automatically translate into an equivalent cost or energy advantage without a fuller life-cycle costing exercise [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025]. More importantly, there is a real durability gap between metal and polymer AM routes: metal PBF mold inserts are reported to reach over 100,000 injection cycles, comparable to conventional tool steel, while polymer rapid-tooling AM inserts can have service lives as low as 1.3% of metal equivalents. That means the material-saving conclusions drawn here apply most reliably to metal AM routes, and that polymer AM, while genuinely useful for prototyping or short-run tooling, is not currently a like-for-like substitute for production-grade CNC mold inserts.

Bringing this back to the real-world scenario from the introduction: the tool shop discarding drums of oily swarf every week is not just facing a scrap-cost problem, it is facing a compounding chain of coolant purchase, filtration, cleaning, and disposal that this study's numbers suggest could shrink by roughly 90% for post-processing effort alone if that shop shifted deep-cavity mold work to metal AM at low infill. That is the kind of concrete, operational translation the sustainability literature has been gesturing toward without always spelling out in kilograms and cycle counts [Alam & Putri, World Journal of Advanced Research and Reviews, 2025].

9. Limitations of the Study

This study carries a few limitations worth being upfront about. First, it is literature-based and quantitative-comparative rather than experimental; no physical CNC machining or AM print trials were run, so every figure here depends on how accurate and comparable the underlying 2025-2026 sources are rather than on direct measurement. Second, the case study models a single representative geometry, a 500 mL bottle-mold cavity insert, and the results, especially the 65.0%

CNC scrap fraction, may not generalize to molds with shallower cavities or simpler geometries where buy-to-fly ratios would sit lower. Third, the combined topology-optimization-plus-infill figures (88-93% total reduction) are estimates built by applying literature-reported percentage ranges to the case-study baseline rather than outputs of an actual topology-optimization simulation, so they should be read as indicative rather than exact. Fourth, the analysis focuses on material mass and does not fold in full cost or energy life-cycle accounting, so it cannot yet confirm whether AM's material savings translate into equivalent economic or energy savings once feedstock price and build-time energy are brought into the picture.

10. Future Directions

Future work should put primary experimental validation first, actually machining and printing matched mold-insert geometries to measure scrap mass, buy-to-fly ratio, and AM material consumption directly rather than relying only on literature synthesis. A dedicated energy-consumption analysis, covering machining power draw, AM build energy, and post-processing energy for both routes, would let the material-saving figures reported here be converted into a full life-cycle cost and carbon comparison. Extending the case study across multiple mold geometries, shallow cavities, multi-cavity molds, and molds with conformal cooling, would test how far the 65.0% CNC scrap finding generalizes beyond the single bottle-mold case used here. Finally, running an actual topology-optimization simulation on the case-study geometry, instead of applying literature-reported percentage ranges, would turn the estimated 88-93% combined-strategy savings into a verified, geometry-specific result.

11. Conclusion

This study set out to quantify how infill-optimized additive manufacturing compares with CNC machining in mold-cavity insert production, using a representative 500 mL bottle-mold case study. The results show that CNC machining of this insert consumes 35.32 kg of P20 billet to retain only 12.36 kg in the finished part, a 65.0% scrap fraction, while the equivalent AM build cuts total material to somewhere between 14.22 kg (100% infill) and 3.34 kg (10% infill), and to as little as 2.5-4.2 kg once topology optimization is added, representing reductions of 60-93% relative to CNC. These savings scale linearly with production volume and are backed up by a much higher multi-criteria sustainability score for AM (8.75/10 versus 2.9/10 for CNC). Infill-density and topology optimization together represent a substantial, quantifiable, and still underused lever for cutting material consumption in mold manufacturing, with the caveat that the strongest conclusions apply to metal AM routes rather than polymer rapid tooling, given how much shorter the latter's service life tends to be.

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Appendix

Table A1. AM Infill Pattern Properties Summary

Pattern	Key Property	Material Use	Best Suited For
Grid	Most consistent tensile/hardness/impact performance	Higher than others at same density	Load-bearing regions
Honeycomb	Curved cell walls trade peak strength for mass saving	Up to 30% less than grid	Weight-sensitive, moderate-load regions
Gyroid	Near-isotropic behavior; excellent strength-to-weight	Moderate; 5-15% slower print speed	Multi-directional loading
Triangle	High shear resistance via diagonal load paths	Between grid and honeycomb	Shear-dominated regions
Concentric	Follows perimeter contours; good surface finish	Low rigidity in thick sections	Thin-walled or flexible parts

Source: [El Omari et al., ISTE OpenScience, 2024] [Raise3D, Raise3D Blog, 2025] [Sovol3D, Sovol3D Blog, 2026] [Xometry, Xometry Resources, 2026].

Table A2. Multi-Factor Sustainability Index (0-10 Scale)

Criterion	CNC Machining	Metal AM (PBF/DED)
Material efficiency	2.5	8.5
Scrap recyclability	5.0	9.0
Energy for scrap recovery	2.5	8.0
Coolant / contamination burden	1.5	9.5
Post-processing effort	2.0	9.0
Circular economy potential	4.0	8.5
Overall score	**2.9**	**8.75**

Source: [Frigate Manufacturing, Frigate.ai, 2025] [Javed et al., International Journal of Advanced Manufacturing Technology, 2025] [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025].

Table A3. CNC vs. AM Scrap Post-Processing Flow

Step	CNC Machining	Metal AM (PBF/DED)
1	Billet	Powder
2	Machining	Printing
3	Metal chips	Unused powder
4	Coolant removal	Sieving
5	Cleaning and drying	Reuse
6	Recycling	-

CNC requires 6 process steps with a high energy and CO2 burden; AM requires 4 steps with a comparatively low energy and CO2 burden (source: [Met3DP, Met3DP Blog, 2025] [Javed et al., International Journal of Advanced Manufacturing Technology, 2025] [Segovia-Guerrero et al., Journal of Manufacturing and Materials Processing, 2025]).

Tools Used

The preparation of this dissertation involved the use of several digital and AI-assisted tools to support data analysis, numerical comparison, formatting, grammar checking, and language refinement. These tools were strictly supportive, and all technical interpretation, validation, and conclusions were performed by the author, ensuring academic integrity.

AI-Based Tools

AI Chatbots (e.g., ChatGPT):

- Used to cross-check numerical consistency across tables, such as lead times, material waste percentages, and cost estimations.
- Assisted in summarizing literature trends, structuring sections, and improving technical clarity.
- All quantitative conclusions were verified manually against original sources to ensure accuracy.

Numerical Comparison Tools

Spreadsheet Software (e.g., Microsoft Excel):

- Organized quantitative data from multiple sources.
- Calculated percentage reductions, cost-per-part estimations, material utilization, and energy consumption.
- Generated tables and figures presented in Sections 4.2 (Temporal Efficiency), 4.3 (Material Waste), and 4.5 (Economic Considerations).

Language and Formatting Tools

1. Grammar and Spelling Checkers (e.g., Grammarly):
   - Improved grammar, sentence structure, and academic tone without altering technical meaning.

2. Paraphrasing and Rewriting Tools:
    - Used selectively to enhance clarity and reduce redundancy while preserving original technical content.
3. Formatting Tools:
    - Applied for consistent headings, table layouts, figure captions, and reference styling according to academic guidelines.

*All tools were applied under **strict academic integrity**, with the author retaining full responsibility for all interpretations, analyses, and conclusions.*